

Introduction to Artificial Intelligence

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines.

AI is used in healthcare, education, finance, and transportation.

The goal is augmentation, not just automation.

Key Concepts in AI

Machine Learning (ML): Systems learn from data

Natural Language Processing (NLP): Understanding human language

Computer Vision: Interpreting images and videos

Robotics: Physical AI systems

Applications of AI

AI enables automation and predictive insights.

Used in healthcare, education, and finance.

Raises ethical concerns like bias and privacy.

Scanned Style Content (Simulated)

Artificial Intelligence systems rely on training data.

Poor data leads to poor performance.

Factors: accuracy, consistency, completeness.

Low Quality Content

AI models need tuning.

Hyperparameters: learning rate, batch size, epochs.

Noise affects OCR accuracy.

Table Content

Model Type	Use Case	Complexity
Linear	Prediction	Low
CNN	Image Analysis	Medium
Transformer	NLP Tasks	High

Mixed Content

AI systems can fail if:

- Data is biased
- Models are overfitted
- Inputs are unexpected

AI Workflow

1. Data Collection
2. Data Preprocessing
3. Model Training
4. Evaluation
5. Deployment

Advantages of AI

- Automation of repetitive tasks
- Faster decision making
- Improved accuracy

Benefit	Description
Speed	Processes large data fast
Efficiency	Reduces manual work
Scalability	Works across systems

Noisy Page

Al... syst3ms sometim3s faiL due t0 err0rs.

misspelled w0rds ### !!!

inc0nsistent f0rmatting